

Mathematics A

Unit 1 • Chapter OL-T44 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

6 classified items • 28 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Dr Eslam Ahmed

Elite IGCSE Mathematics • eliteigcse.com

WhatsApp +20 112 000 9622

ELITE TEACHING EDITION - REVIEWED FOR WEBSITE USE

Contents

Unit 1 • Expertise • Unit 1 • Chapter OL-T44 • Expertise Q16+ • 6 items • 28 marks

Chapter	Items	Marks	Starts
01. OL-T44 Circles, Arcs & Sectors	6	28	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Circles, Arcs & Sectors

Unit 1 • Expertise • 28 marks • 6 item(s)

November 2023 | 4MA1-2H Q20 | 4 marks

Find annular, overlapping or multi-circle area

June 2023 | 4MA1-1H Q22 | 6 marks

Solve an algebraic circle-measure constraint

November 2024 | 4MA1-1H Q16 | 4 marks

Find sector area

November 2024 | 4MA1-2H Q25 | 5 marks

Find annular, overlapping or multi-circle area

January 2022 | 4MA1-1H Q16 | 4 marks

Recover radius or central angle from an arc/sector measure

January 2023 | 1H Q21 | 5 marks

Find a circular segment area or perimeter

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-2H Q20 whole

4 marks

- 20 The diagram shows equilateral triangle ABC with sides of length 10 cm.
A circle is drawn inside the triangle.

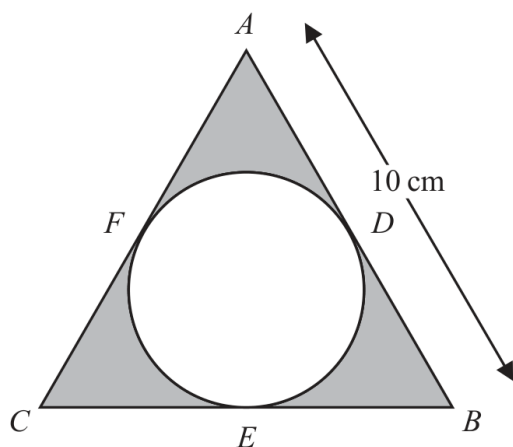


Diagram **NOT**
accurately drawn

D , E and F are points on the circle.

ADB , BEC and CFA are tangents to the circle.

Calculate the total area of the regions shown shaded in the diagram.

Give your answer correct to 3 significant figures.

..... cm^2

WORKING SPACE

Continue your method**4 marks**

WORKING SPACE

4MA1-1H Q22

6 marks

22 The diagram shows two circles with centre O and a regular pentagon $ABCDE$

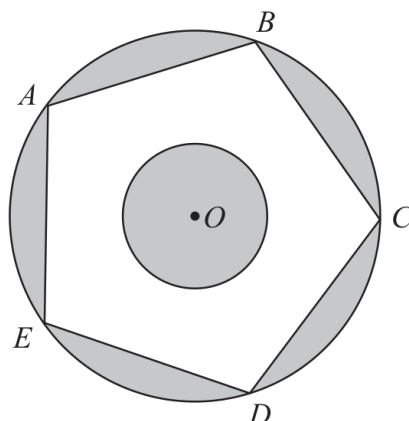


Diagram **NOT**
accurately drawn

A , B , C , D and E are points on the larger circle.
The pentagon has sides of length 8 cm.

The diagram is shaded such that

$$\text{shaded area} = \text{unshaded area}$$

Work out the radius of the smaller circle.
Give your answer correct to 3 significant figures.

..... cm

Continue your method

6 marks

WORKING SPACE

4MA1-1H Q16 M01

4 marks

16 $OAPB$ is a sector of a circle, centre O

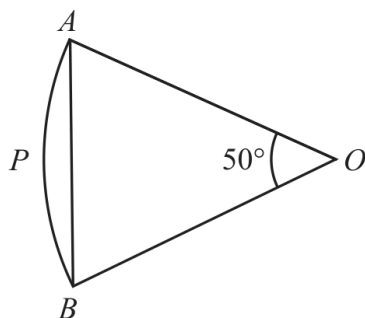


Diagram **NOT**
accurately drawn

Angle $AOB = 50^\circ$

Area of triangle $OAB = 120 \text{ cm}^2$

Work out the area of the sector $OAPB$

Give your answer correct to 3 significant figures.

..... cm^2

WORKING SPACE

4MA1-2H Q25 M01

5 marks

25 The diagram shows an equilateral triangle ABC and a circle with centre O

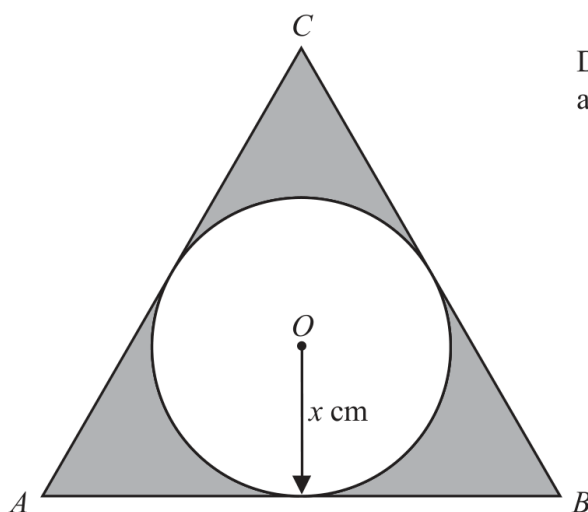


Diagram **NOT**
accurately drawn

AB , BC and CA are tangents to the circle.

The radius of the circle is x cm

The total area, in cm^2 , of the regions shown shaded in the diagram is nx^2

Find the value of n

Give your answer correct to 3 significant figures.

4MA1-2H Q25 M01

5 marks

 $n = \dots\dots\dots$

WORKING SPACE

4MA1-1H Q16

4 marks

16 The diagram shows a circle with centre O

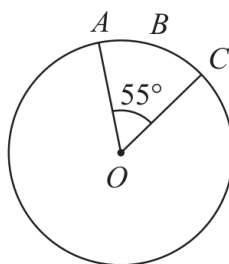


Diagram **NOT**
accurately drawn

A , B and C are points on the circle so that the length of the arc ABC is 5 cm.

Given that angle $AOC = 55^\circ$

work out the area of the circle.

Give your answer correct to one decimal place.

..... cm^2

Continue your method**4 marks**

WORKING SPACE

21 The diagram shows the cross section of a circular water pipe.

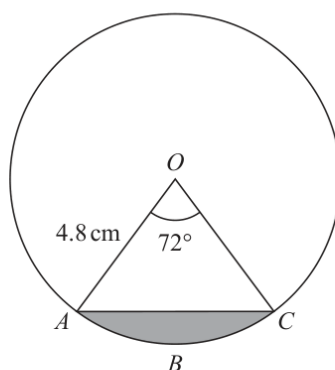


Diagram **NOT**
accurately drawn

OAC is a sector of the circle, centre O

The shaded region in the diagram represents the water flowing in the pipe.

The water flows at 14 cm/s in the pipe.

Work out the volume of water that has flowed through the pipe in 3 minutes.
Give your answer in cm^3 correct to 3 significant figures.



YOUR WORKING

..... cm³

(Total for Question 21 is 5 marks)

YOUR WORKING
